Recovery trials, explained

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Three projects, one exercise, and what the first trial can and cannot tell us.

Who this is for

Anyone who lands on this repository without having followed the threads on the Information Physics Institute list. It explains what Observer Patch Holography, F O R M and MORPH are, why they fit together, how a recovery trial works, and where the first trial stands. The rules live in PROTOCOL.md and in the card of each trial; this page is the plain version.

Three projects

Observer Patch Holography: a theory that runs

Observer Patch Holography (OPH) is Bernhard Mueller's formal theory of physical reality, built to be consistent, complete and executable. Its starting point is the observer. An observer keeps records, repairs the damage that each new interaction does to them, and remembers. The theory has one local law for that repair and memory, acting on small units called twelve-port carriers, and everything else is derived from it. Physics, in this picture, is what observers jointly recover from the records they share.

Two things make OPH unusual. The derivations are machine-checked: more than 12,700 theorems in the Lean proof assistant, with no admitted proofs. And there is an executable model: code anyone can run, which produces records, reads them back and prints the numbers. Among the results proved on their stated domain or executed with retained receipts are the Born rule, the Tsirelson bound, a slow response with exactly three spatial directions, the gauge Lie algebra of the Standard Model, a 1+3 causal order with proper time read from record counts, the Einstein tensor from null balance and conservation, and dark energy at (w0, wa) = (−1, 0).

What OPH does not have is particles. A particle in OPH is a defect that local repair cannot undo, and its mass is its upkeep. Several candidate mass laws have been tried and none reproduces the measured masses. The one that matters for this repository gave the three charged leptons in the ratio 1 : 1.19 : 1.50; the measured ratio is 1 : 206.8 : 3477. The law and its failure are both public in the OPH repository, and that is by design. Condition 4 of the OPH mission is agreement with measurement, with predictions frozen before the test.

F O R M: what a reader is entitled to recover

F O R M is Stefaan Vossen's instruction for reading scientific objects. Its motto: FORM the object. FORMulate the relation. FORMalise only the standing it survives. A proof assistant starts from declared premises and asks what follows. F O R M starts from whatever representation is available, a paper, a dataset, a corpus, a set of records, and asks what a reader may legitimately recover from it, and with what standing.

Its working parts are labels. Every component of a reading says where it entered the record: SOURCE-STATED (the source says it), SOURCE-IMPLIED, RECOVERED (the reader derived it from what the source supplies) or EVALUATOR-INFERRED (the reader brought it in). A reading as a whole gets a disposition: CONVERGENT when it agrees with the source and keeps its distinctions, OVER-RECOVERED when it attributes structure the source does not support, UNRESOLVED when nothing more specific is justified, and so on. Two rules matter most here. Independent readers agreeing with

each other is a different thing from any of them being right about the source. And the instruction applies to itself: its own provenance and drafting history have to be recoverable in the same way.

F O R M also carries Diana Haskins's stop rule, FER. If establishing a correspondence would need a bridge the sources do not state, stop there and record the stop instead of building the bridge.

The closing instruction of F O R M v1.0 says what it is for: "It asks what happened to the object, under what conditions, what survived the operation, how that survivor is represented, what standing was actually earned and what another operator is entitled to do next."

MORPH: the transformation and its scar

MORPH (Stefaan writes it M⊙RPH) is the youngest of the three. In his words it is less a new theory than an application successor to F O R M for settings where information is limited and something gets transformed on the way. It names the five stations of such a transformation:

- M, the model: the representation actually available;
- ⊙, the operator: the transformation actually performed;
- R, the reconstruction: what the surviving record licenses us to recover;
- PH, patch healing: a successor object in which the effects of the transformation, the losses and the corrections stay recoverable instead of being erased;
- M_1, the successor model.

The letters PH first stood for Patch-Holographic, with OPH as the mathematical example at hand; in October Stefaan re-read them as Patch Healing, keeping OPH as the reference. The healing reading comes with an image this repository takes seriously. A healed object has passed through an event, and the event becomes part of its history. If something was misattributed and later corrected, the correction does not erase the misattribution; both belong to the successor. That is the scar.

MORPH also supplies five words for comparing what two sides hold, transition by transition: PRESERVED, CORRESPONDENT, SOURCE-SPECIFIC, UNRESOLVED and INCOMPATIBLE. And from Stefaan's Constitutional Incarnation work comes a ladder that suits an executable theory. An illustration depicts a rule. An incarnation behaves according to the rule, so that the rule can be recovered from what happens. Validation needs its own, independently justified conditions. Incarnation is stronger than illustration and weaker than validation.

Why the three fit together

All three are about records and what they license, from different sides. OPH says the world is records and physics is what observers jointly recover from them. F O R M says a reading is entitled to what the representation licenses, and no more. MORPH says what happens to a record on its way from one holder to the next, and insists that the losses and corrections stay visible. Put the three in one room and the same question arrives from three directions: what does this record carry?

Each supplies what the others lack.

- OPH gives F O R M an ideal test object. F O R M's usual objects are papers and corpora, where nobody holds a key that says what the author meant. OPH's executable model produces records from a fully declared instrument, so the origin of every record can be written down and sealed before anyone reads. For once, a reading can be scored against the truth.
- F O R M gives OPH a discipline for the step OPH got wrong in public. The lepton law passed every formal step and failed at the last one, the sentence "these three numbers are the electron, muon and tau". No record contains that sentence. F O R M has a name for inferences of that kind and a rule for withholding them.

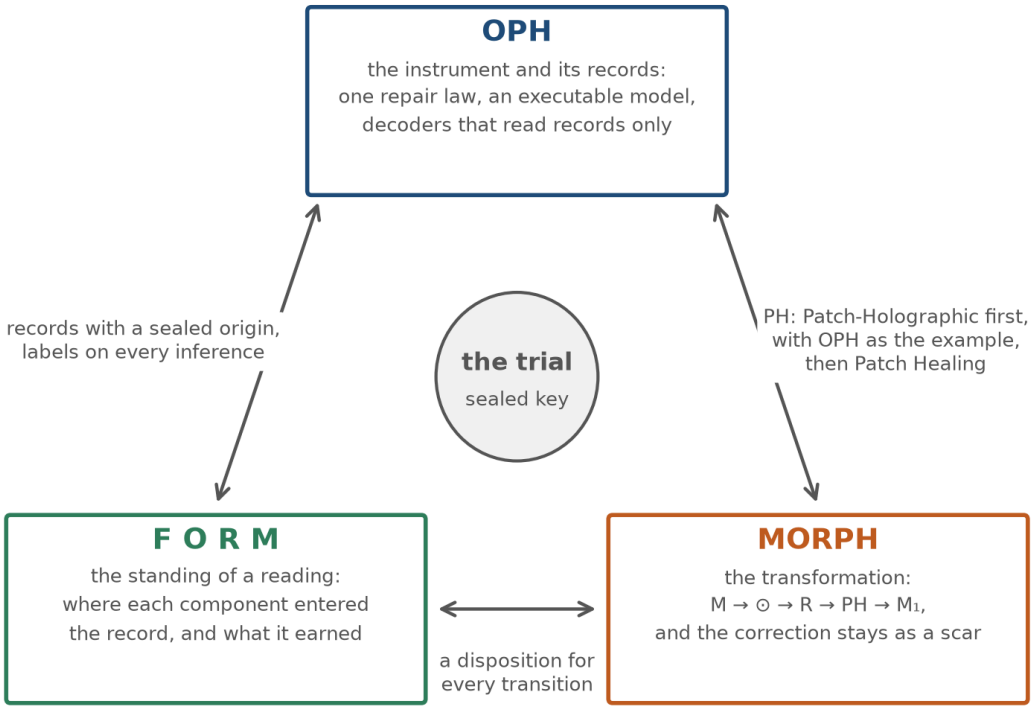


Figure 1: The three projects and the trial between them

- MORPH gives the exercise its shape. The whole transformation is recorded, from the records as published to the scored successor object, and the wrong readings stay in it as scars. The lepton law's failure is the first scar, kept on purpose.
- The directions complete each other. OPH's Lean library runs forward, from premises to consequences. F O R M runs backward, from a representation to what it entitles. OPH's executable model has decoders that run backward as well, reading records with no access to the law that produced them. A trial puts the two backward readers, machine and human, on the same records.

None of this validates any of the three. Physical validity stays with measurement. A trial tests one thing: whether the step from records to names is licensed by the records or added by the reader.

How a trial works

1. M. The trial authors publish the record streams with their identities removed and their order shuffled, together with a one-page card: what the instrument is, what each bit means, which facts the source states, what to recover, and how it will be scored. The key (which stream is what, how the constructed streams were made, the random seed) is written down and hashed, the hash is

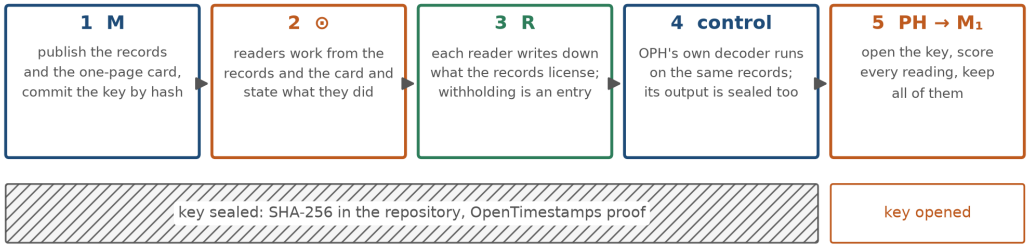


Figure 2: The five steps of a trial

published, and an OpenTimestamps proof anchors the hash in the Bitcoin blockchain, so the date cannot be moved later.

2. ⊙. Readers work from the records and the card alone and say what they did. Any tool is allowed, including the instrument's own code.
3. R. Each reader writes down what the records entitle them to recover, with a provenance label on every component and a disposition on the whole. Withholding is an entry. A reader who meets an unstated bridge stops and records the stop.
4. Control. The authors' own decoder reads the same records. Its output is sealed with the key, so it can be scored like everyone else's.
5. PH → M_1. After the deadline the key is opened and a public script scores every reading against it. Everything is kept: the right readings, the wrong ones, the withheld ones, the control, the key, and the authors' own mistakes.

What the seal fixes

Before anyone reads, three files are written and hashed, and the hashes are published with a timestamp proof. Together they fix:

- The answer sheet. For every stream: its role (the published law, or constructed and how), the exact coefficients that produced its table, the three gaps a correct reading should find, the coefficient arguments the records do not carry, and the dictionary entry OPH holds for it, a particle name or none.
- The construction. The script that built the key, the random seed and the rule that produced it. After unsealing, anyone reruns the script and the generator and gets the same tables.
- The control. What OPH's own decoder returned on the six streams, so OPH's performance is fixed before readers start, like everyone else's.
- The published material. A second hash list covers the card, the streams, the tools and the protocol as they stood at sealing, so the rules cannot drift while readings come in.
- The instrument. The commit of the OPH release and the hashes of the files that define the instrument, the decoder and OPH's statements about the published law.

The hash binds: one changed byte in any of these files changes it, so the key cannot be adjusted after submissions arrive. The timestamp dates: the hash sits in a Bitcoin block, so the seal cannot be moved later. Neither hides the numbers. The tables are public, and a reader with the decoder can read all six. What stays sealed is the authoritative account of where each table came from and what OPH holds about it, which is what a reading is scored against.

What the scores mean

Each component of each reading gets one of the five MORPH words against the key. PRESERVED: they agree within the tolerance on the card. CORRESPONDENT: close, short of the tolerance. SOURCE-SPECIFIC: the reading reports something the key holds and the records do not carry, or the reverse. UNRESOLVED: withheld or not settled. INCOMPATIBLE: they disagree. Where a reading attributes structure the key does not hold, the F O R M label OVER-RECOVERED is added. Agreement among readers and agreement with the key are reported as separate numbers, and a majority of readers agreeing on something the key does not hold is recorded as CONSENSUAL OVER-RECOVERY.

What a trial does not establish

Recovering the sealed dictionary shows that readers matched a declared reference. It does not show that the dictionary is uniquely recoverable from the records; that needs an argument of its own, and reader agreement cannot supply it. And a trial says nothing about whether OPH's physics is right. That is decided by measurement.

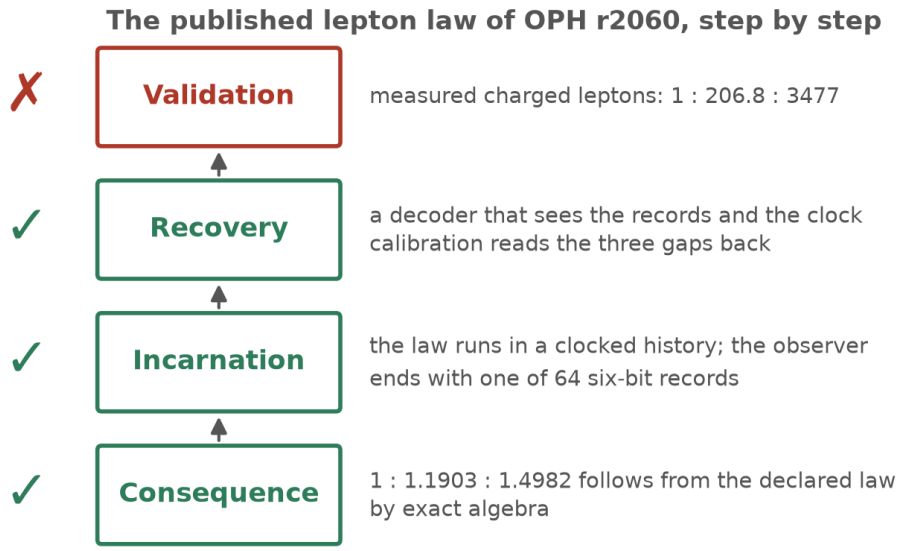


Figure 3: The published law, step by step

Trial 01: the lepton negative control

A negative control is a case where the right answer is known in advance, run to check that a procedure returns it. Here the known answer is the failed lepton law.

The law says: the three families of charged leptons are three directions of OPH's slow response band, a prepared reference state breaks the symmetry between them, and the masses are fixed multiples of the resulting eigenvalues. From that, by exact algebra, the three gaps come out in the ratio 1 : 1.1903 : 1.4982. The law runs as a clocked history in the executable model; an observer who watches it ends with one of 64 six-bit records. A decoder that sees the records and the clock calibration reads the three gaps back to nine decimal places. Then the comparison with measurement: 1 : 206.8 : 3477. The law fails, and a proved bound shows that no law of its form can reach the measured hierarchy.

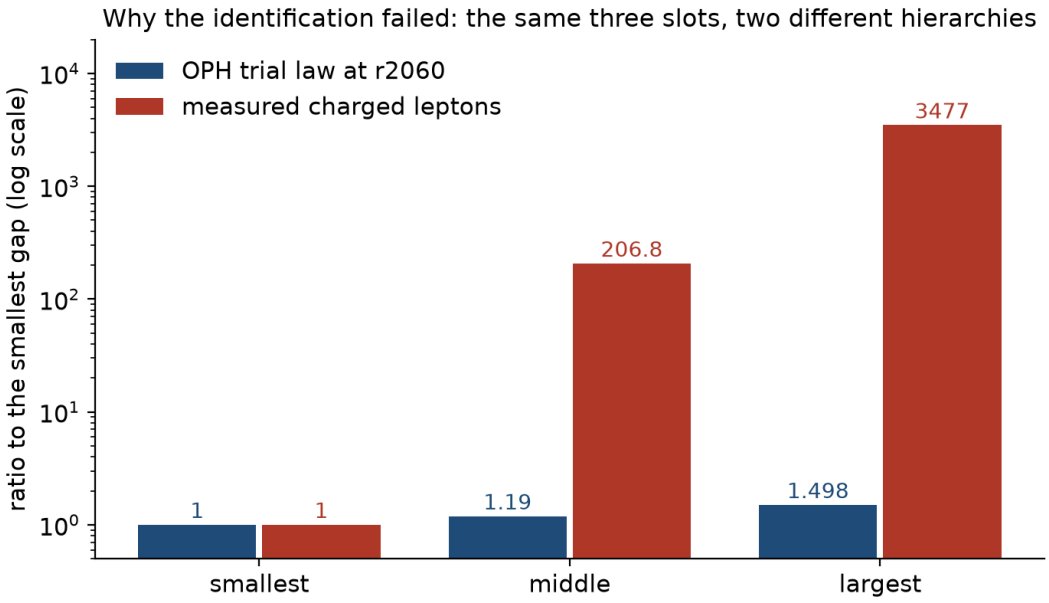


Figure 4: Two hierarchies in the same three slots

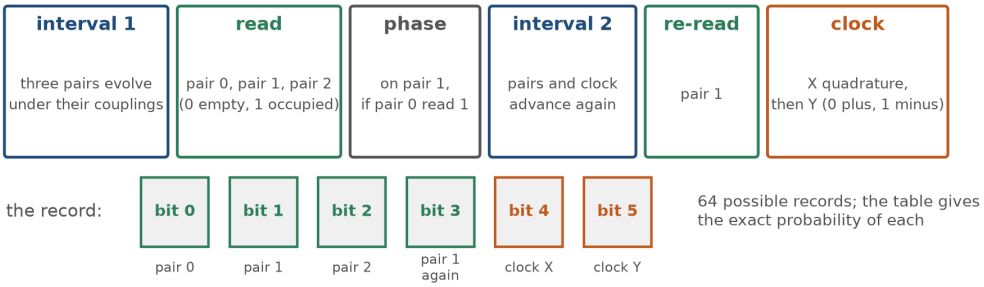


Figure 5: One run of the instrument and the six-bit record it leaves

Three steps survived and one failed. The step that failed was an interpretation carried forward: “these three directions are the electron, muon and tau”. That sentence is in no record. Trial 01 asks whether independent readers, given the records and the card, recover the numbers and decline the sentence.

The object

The instrument is public code in OPH release r2060. Three pairs of fermion modes start empty and evolve under a pairing coupling for one interval; each pair is read; a conditional phase is applied; a second interval runs; one pair is read again; and a two-qubit clock that advanced through both intervals is read in two quadratures. The six readings are the record. The table of a stream lists all 64 possible records with their exact probabilities; nothing is sampled.

Six such tables are published. One is the published law, run as published. The other five were produced by the same instrument from coefficients the trial authors chose; the card says so and warns that some may coincide with values a reader recognises. Which is which is sealed.

For each stream a reader reports the three gaps in clock units, an origin (the published law, unknown, or something else), a particle name for each slot or WITHHELD, the provenance of each, a disposition, and any stop point. The scoring script is public, so a reader can see in advance how each field is scored.

What is known at sealing

- Protocol 1.0 and trial 01 were frozen and sealed on 3 November 2026. The hashes of the key, the control output and the key script are in the trial folder with their timestamp proofs.
- The published law’s record table regenerates from the pinned OPH code and matches the receipt in the OPH repository to one part in 10^14.
- OPH’s own decoder, run on the six published streams, recovers every sealed value within three parts in a million. That is the control; readers can do better or worse.

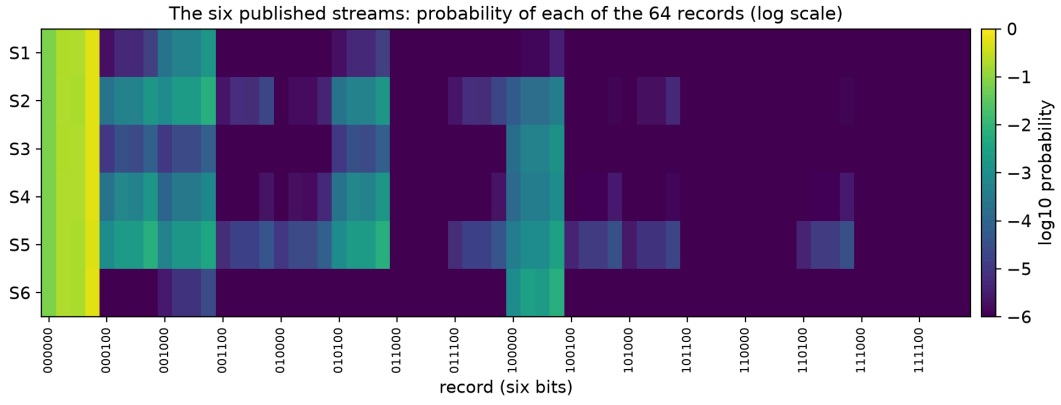


Figure 6: The six published streams

- The records carry only the size of each coupling. Its complex phase leaves the table untouched, so a reader who reports one has read in more than the records hold. That is a built-in case of the SOURCE-SPECIFIC disposition.
- Exact tables are needed. From sampled records, gaps below about a thousandth of the clock cannot be recovered at any practical sample size.

What happens next

The deadline is set on the Information Physics Institute list once the protocol has been read. Readings arrive as pull requests. At the deadline the key is opened, the script scores everything, and the scores, the key and every reading go into the repository together. If a later OPH mass law gives the measured masses, a second trial can ask the harder question: whether independent readers recover a dictionary from defects to particles that OPH has sealed in advance.

Glossary

term	meaning here
record	the six bits an observer ends up with after one run of the instrument
stream	the table of all 64 records of one run and their probabilities
card	the one page a reader works from
key	the sealed file that says which stream is what
control	OPH's own decoder, run on the published streams, output sealed
gap in clock units	the quantity to recover for each pair: its coupling strength relative to the clock's frequency
negative control	a case whose right answer is known, run to check the procedure
SOURCE-STATED, SOURCE-IMPLIED, RECOVERED, EVALUATOR-INFERRED	F O R M provenance labels: where a component entered the record
CONVERGENT, PARTIAL, NON-UNIQUE, UNDER-RECOVERED, OVER-RECOVERED, SEMANTIC DISPLACEMENT, INCOMPATIBLE, UNRECONSTRUCTABLE, JURISDICTION DRIFT, CONSENSUAL OVER-RECOVERY, UNRESOLVED	F O R M dispositions of a reading
PRESERVED, CORRESPONDENT, SOURCE-SPECIFIC, UNRESOLVED, INCOMPATIBLE	MORPH comparison words, used by the scorer against the key
FER	the stop rule: at an unstated bridge, stop and record the stop
scar	a correction kept next to the error it corrects

Links

- This repository: https://github.com/muellerberndt/recovery-trials
- The protocol: PROTOCOL.md

- Trial 01: trials/01-lepton-negative-control
- Observer Patch Holography: https://github.com/FloatingPragma/observer-patch-holography
- F O R M v1.0: https://zenodo.org/records/22875795
- Constitutional Incarnation: https://www.dottheory.co.uk/paper/incarnation